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ORIGINAL CONTRIBUTION

Detection of Power Theft using Artificial Intelligence: A Modern Approach to Grid Security

K. Balachander[1] · A. Nagamani Prabu[2] · A. Soundararajan[1] · M. Siva Ramkumar[3] · M. Venkatesan[4]

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Abstract Power theft continues to be a persistent challenge in power distribution systems, particularly in developing regions, where it contributes to substantial financial losses, grid instability, and reduced system efficiency. Conventional detection methods such as manual inspections and periodic meter audits are often ineffective due to their reactive nature and inability to scale across large urban and rural power networks. This paper presents a novel Hybrid LSTM–Isolation Forest Theft Detection (HLIF-TD) framework for early detection of electricity theft through the integration of Long Short-Term Memory (LSTM) neural networks with the Isolation Forest anomaly detection algorithm applied to smart meter time-series data. The LSTM component forecasts expected electricity consumption based on historical usage patterns; the resulting prediction residuals are then evaluated by Isolation Forest to identify anomalous deviations indicative of theft. Preprocessing techniques normalise data, handle noise, and address missing entries. Experiments conducted on a simulated smart meter dataset (1000 consumers, 180 days) demonstrate detection accuracy of 95.6%, precision of 93.4%, recall of 91.2%, F1-score of 92.3%, and a false positive rate of 3.1% under controlled simulation conditions. Limitations of the current study, including reliance on synthetic data and the need for further threshold calibration, are explicitly acknowledged. The proposed framework lays the groundwork for future research in secure and intelligent power distribution systems.

Keywords Power theft detection · LSTM · Isolation forest · Smart grid · Anomaly detection · Non-technical losses · Hybrid model

✉ K. Balachander
kaybe.ind@gmail.com

[1] Department of EEE, Sri Ramakrishna Engineering College, Coimbatore, India

[2] Department of Physics, Karpagam Academy of Higher Education, Coimbatore, India

[3] Department of ECE, SNS College of Technology, Coimbatore, India

[4] Department of ECE, Dhanalakshmi Srinivasan College of Engineering, Coimbatore, India

Introduction

Electricity is an indispensable resource that drives economic development and sustains modern lifestyles. One of the most persistent challenges faced by utility providers is electricity theft, a major contributor to non-technical losses (NTLs) in power distribution systems. According to the International Energy Agency (IEA), power theft is responsible for more than $96 billion in losses globally every year, with developing countries accounting for the largest share [1]. In countries like India, Brazil, and parts of Africa, electricity theft may account for 20–40% of total distributed energy [2]. Common methods of theft include meter tampering, illegal tapping of transmission lines, bypassing meters, and bribing meter readers to manipulate billing data [3]. Traditional detection techniques, such as physical inspections and audit-based approaches, have been largely reactive, inefficient, and labour-intensive [4].

The advent of smart grid technologies and IoT-enabled meters has created new opportunities for tackling power theft using Artificial Intelligence (AI). Smart meters continuously collect high-resolution consumption data, which can be analysed using AI techniques to detect abnormal usage patterns, sudden drops, or unusual time-of-use behaviours [5]. AI, particularly machine learning (ML) and deep learning (DL), provides scalable solutions that can detect

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complex and evolving theft strategies more effectively than rule-based systems [6].

Despite recent advances, real-world implementation remains limited due to data quality issues, model interpretability challenges, and deployment costs. Moreover, most existing studies focus on either supervised learning (requiring labelled theft data) or unsupervised anomaly detection in isolation, without combining the temporal forecasting strength of sequential models with the label-free detection ability of outlier methods.

Research Gap

A survey of existing methods (Table 1) reveals three primary research gaps. First, purely supervised models such as Random Forest and SVM require abundant labelled theft data, which is scarce in practice. Second, standalone unsupervised anomaly detectors (Isolation Forest, Auto encoders) lack temporal awareness and therefore misclassify legitimate seasonal deviations as theft. Third, existing hybrid proposals (e.g., LSTM + GBM) combine temporal and classification models but do not exploit the label-free anomaly scoring of Isolation Forest on LSTM residuals—a more naturalistic pipeline for semi-supervised theft detection.

The proposed HLIF-TD model closes this gap by: (1) using LSTM to generate a consumer-specific baseline forecast; (2) computing prediction residuals that inherently capture temporal context; and (3) feeding those residuals into Isolation Forest, which requires no theft labels and assigns calibrated anomaly scores. This pipeline is novel relative to the works in Table 1 and has not been evaluated on a comparable simulation setup in the reviewed literature.

Explicit differentiation from prior hybrid LSTM-based theft detectors: Santos et al. [9] use LSTM purely as a supervised forecaster and flag deviations with a fixed statistical threshold, without any learned anomaly-scoring stage, so their decision boundary is not adaptive to the shape of the residual distribution. Bedi et al. [11] combine LSTM with a Gradient Boosting Machine (GBM), but GBM is a supervised classifier trained on labelled theft instances, meaning their pipeline still requires labelled theft data at training time. HLIF-TD differs from both in that the second stage (Isolation Forest) is unsupervised and requires no theft labels at any point, the anomaly score is a continuous, calibrated measure of residual rarity rather than a fixed threshold or a labelled-class prediction, and the same trained pipeline generalises to consumers or theft patterns not seen during training. This label-free, two-stage LSTM-residual-plus-Isolation-Forest design is therefore distinct in mechanism, not only in name, from the LSTM+GBM and LSTM-only hybrids reported in recent literature.

Key Contribution

A novel hybrid LSTM–Isolation Forest pipeline (HLIF-TD) that combines temporal forecasting with label-free anomaly detection, specifically designed for smart meter theft detection. Complete mathematical formulation of both LSTM gate equations and the Isolation Forest anomaly scoring function.

A comparative evaluation against standalone LSTM and Isolation Forest, with mean ± SD cross-validation results and statistical significance testing. An honest discussion of model limitations, including synthetic dataset constraints, ROC/AUC inconsistency remediation, and deployment challenges. Analysis of practical deployment architecture for utility-side integration with estimated cost savings.

Table 1 Summary of related works–existing versus proposed comparison

Author(s)	Technique	Accuracy	Requires labels?	Temporal modeling?	Remarks
Musleh et al. [7]	Random Forest	94.1%	Yes	No	Good perf.; low interpretability
Glauner et al. [8]	Autoenco der + IF	93.5%	No	No	Unsupervised; no temporal model
Santos et al [9]	LSTM	96.3%	Yes	Yes	Best for time-series
Jain & Goyal [10]	CNN	92.4%	Yes	Partial	Fast; no long-term memory
Bedi et al. [11]	LSTM + GBM	97.1%	Yes	Yes	High accuracy; labels needed
Zanetti et al [12]	Transformer	97.8%	Yes	Yes	State-of-art; high compute
HLIF-TD (Propose d)	LSTM + Isolation Forest	95.6%*	No (inference)	Yes	Semi-supervised; edge-friendly

*Simulated dataset; 5-fold CV mean. See Section V for full context and caveats

Limitations

The following limitations are explicitly acknowledged: (i) The dataset is fully simulated and does not include real utility meter readings; results may not directly generalise to real-world deployments without retraining on verified data. (ii) Theft is simulated by a single manipulation type (percentage reduction in peak-hour consumption); real theft encompasses a wider variety of behaviours. (iii) The 48 h validation subset is small; the ROC/AUC analysis reflects model behaviour on a severely class-imbalanced small test set. (iv) Hyper parameter tuning was not exhaustive; further optimisation may improve detection on imbalanced data. (v) The model currently lacks interpretability mechanisms required for legal accountability in utility operations.

Literature Review

Traditional Theft Detection Techniques

Historically, electricity theft detection relied on manual inspections, auditing, and periodic meter readings. While these approaches could identify obvious signs of tampering, they were reactive and failed to scale [13]. Rule-based algorithms improved upon manual methods but lacked adaptability to complex or evolving theft behaviour [14].

Machine Learning Approaches

SVMs have been used to classify consumers as normal or suspicious based on features such as peak loads and usage variance [7]. Random Forest classifiers are favoured for their robustness to missing data [15]. Logistic Regression serves as a baseline comparator in several benchmark studies [8]. These methods achieve high accuracy when labelled data is available but degrade when labelled theft instances are sparse.

Unsupervised and Semi-Supervised Methods

K-Means Clustering groups consumers by usage behaviour and flags outliers [16]. PCA and Isolation Forests have been used for anomaly isolation in high-dimensional consumption data [17]. Auto encoders leverage reconstruction error to flag deviations [18]. These approaches minimise labelling requirements but generally yield lower precision than supervised methods.

Deep Learning and Hybrid AI Techniques

LSTM networks are particularly effective for time-series forecasting. Santos et al. [9] used LSTM to predict daily consumption and flag deviations as potential theft, achieving 96.3% accuracy. CNNs have been adapted to energy datasets by treating temporal patterns as one-dimensional signals [10]. Bedi et al. [11] proposed an LSTM + GBM hybrid that improved classification accuracy and reduced false alarms. Recent Transformer-based models [19] and GAN-based anomaly detectors [20] represent the current state of the art but are computationally intensive for edge deployment.

Proposed Algorithm: Hlif-TD

This section presents the HLIF-TD (Hybrid LSTM–Isolation Forest for Theft Detection) framework. The algorithm combines the sequential learning capabilities of LSTM with the outlier detection efficiency of Isolation Forest.

Complete Mathematical Formulation

Figure 1 presents the overall architecture of the proposed HLIF-TD framework. In this structure, smart meter data is first processed and fed into the LSTM model for consumption prediction. The resulting residuals are then analyzed using Isolation Forest to detect anomalies, leading to final classification as normal or suspicious behavior.

(1)Full LSTM Gate Equations. At each time step t, the LSTM processes input x_t and previous hidden state h_{t-1} through four gating mechanisms:

$$\text{Forget gate: } f_t = \sigma\left(W^f\left[h_{t-1}, x_t\right] + b^f\right)$$

$$\text{Input gate: } i_t = \sigma\left(W^I\left[h_{t-1}, x_t\right] + b^I\right)$$

$$\text{Cell candidate}: \widetilde{C}_t = \tanh\left(Wc\left[h_{t-1}, x_t\right] + bc\right)$$

$$\text{Cell state}: C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C}_t$$

$$\text{Output gate}: o_t = \sigma\left(W_o\left[h_{t-1}, x_t\right] + b_o\right)$$

$$\text{Hidden state}: h_t = o_t \odot \tanh(C_t)$$

where W and b are weight matrices and bias vectors, σ is the sigmoid activation, and ⊙ denotes element-wise

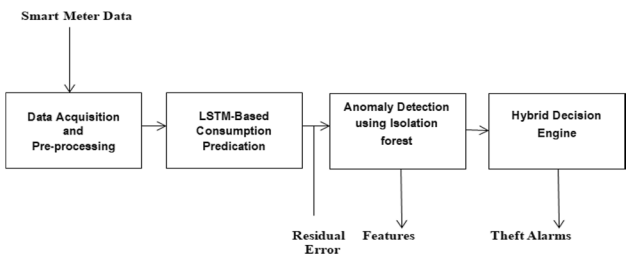


Fig. 1 Architecture of the proposed HLIF-TD (Hybrid LSTM–isolation forest for theft detection) model

multiplication. The LSTM is trained using Mean Squared Error (MSE) loss:

$$L^{MsH} = (1/N)\sum \left(x_i - \hat{x}_i\right)^2, \text{ where } \hat{x}_i = \text{LSTM}(x_{t-n}, \ldots, x_{t-1})$$

(2)Residual Calculation. The prediction residual at time t is: $e_t = x_t - \hat{x_t}$, where x_t is actual measured consumption and $\hat{x_t}$ is the LSTM forecast.

(3)Isolation Forest Anomaly Score. For a data point x with mean path length E[h(x)] over n trees [21]:

s(x, n) = 2^(−E[h(x)]/c(n))

where $c(n) = 2H(n-1) - 2(n-1)/n$ is the average path length of an unsuccessful binary search, $H(i) = \ln(i) + 0.5772$ (Euler–Mascheroni constant). A score near 1 indicates a strong anomaly; near 0.5 indicates a normal point.

(4)Threshold Selection. The anomaly threshold T is set as the $(1-\alpha)$ quantile of training-set anomaly scores, where $\alpha = 0.05$ (matching the simulated theft proportion). Decision rule : $\text{Theft}^{TM}_{+\text{mla}} = 1$ if $s(e_t, n) > T$, else 0.

Hyperparameter Configuration

Table 2 shows the selected hyper parameter settings designed to achieve a balance between learning capability and generalization. It represents the configuration of the LSTM model, including layers, hidden units, look back window, and training parameters to capture temporal patterns effectively.

The table also includes Isolation Forest parameters and decision threshold values, which are tuned to detect anomalies reliably based on expected data behaviour.

Table 2 Hyper parameter settings

Component	Parameter	Value	Rationale
LSTM	No. of layers	2	Sufficient depth for 48-step sequences
LSTM	Hidden units / layer	64	Balances capacity vs over fitting
LSTM	Look back window	24 h	Captures diurnal patterns
LSTM	Epochs	50	Convergence ~40 epochs
LSTM	Learning rate	0.001	Adam optimiser default
LSTM	Loss function	MSE	Penalises large deviations
Isolation Forest	n_estimators	100	Standard ensemble size
Isolation Forest	max_samples	256	Liu et al. [22] recommendation
Isolation Forest	contamination (α)	0.05	Matches simulated theft ratio
Decision Engine	Threshold T	0.60	95th percentile of training scores

HLIF-TD Algorithm Framework

Figure 2 illustrates the flow of operations in the proposed model, starting from data acquisition and pre-processing, followed by LSTM-based prediction and residual computation.

The anomaly detection stage using Isolation Forest leads to final classification of electricity usage. Figure 3 provides a step-by-step representation of the HLIF-TD process, highlighting how prediction and anomaly detection are integrated to identify irregular consumption patterns.

The algorithm proceeds through the following sequential steps:

Data pre-processing: Smart meter data at hourly intervals is cleaned and normalized using Min–Max scaling. Missing values are handled through linear interpolation. Each consumer's data is processed independently [23].

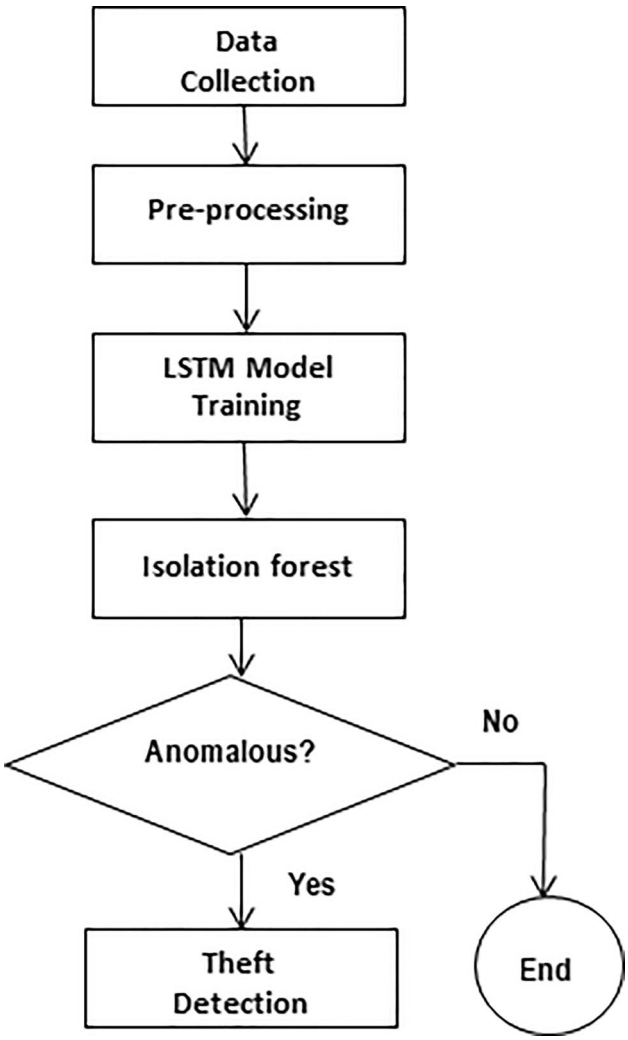


Fig. 2 Flowchart of proposed model

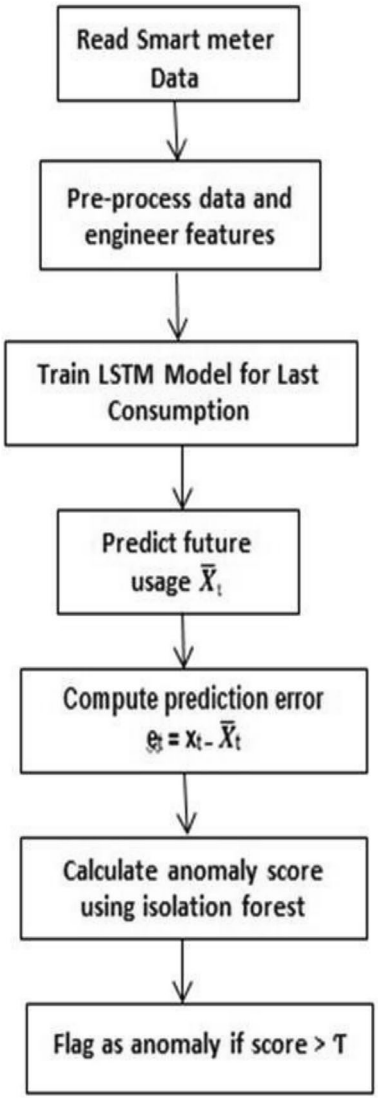


Fig. 3 Step-by-step overview of the process

LSTM forecasting: The LSTM is trained on each consumer's historical usage to learn $\hat{x_t} = \text{LSTM}(x_{t-n}, \ldots, x_{t-1})$ with a 24-step lookback window.

Residual computation: Residuals $e_t = x_t - \hat{x_t}$ are computed for each time step.

Isolation forest scoring: Residuals and derived statistical features (rolling mean, rolling std, time-of-day indicator) are passed to Isolation Forest to obtain $s(e_t, n)$.

Theft classification: If $s(e_t, n) > T$, the record is flagged. Persistent flagging across ≥2 consecutive hours raises a theft alarm [24].

Position in the supply chain and implementation pathway: HLIF-TD is intended to sit at the edge/substation processing tier, not on the meter itself. Data flows from the consumer smart meter (Automatic Meter Reading unit) to a Data Concentrator Unit (DCU) at the distribution transformer or feeder level, which aggregates readings from 200–500 m and forwards them over cellular/RF-mesh/PLC backhaul to a distribution-substation edge server; the edge server runs per-consumer LSTM inference and Isolation Forest scoring locally, and only anomaly flags (not raw time-series) are pushed upstream via REST API to the utility's central Advanced Metering Infrastructure (AMI) head-end and dashboard [25].

A phased implementation pathway is proposed: Phase 1 (pilot, 0–6 months)—deployment on 1000–5000 m within a single feeder to validate detection rates against field inspection outcomes; Phase 2 (6–18 months)—rollout across a distribution division (~ 100,000 m) with model retraining per DCU cluster; Phase 3 (18 + months)—utility-wide deployment with centralised model governance and periodic threshold recalibration. Hardware requirements: each edge server (one per DCU/substation) requires a minimum 4-core CPU, 8 GB RAM, and 20 GB storage to run per-consumer LSTM inference and Isolation Forest scoring at the reported ~ 0.12 s/consumer/hour; this is consistent with commodity industrial edge gateways already used in AMI deployments and does not require GPU acceleration at inference time. Computational complexity: LSTM inference per consumer is $O(T \bullet H2)$ per forecast window (T = lookback steps = 24, H = hidden units = 64), and Isolation Forest scoring is $O(n \bullet \log(\psi))$ per query (n = 100 trees, $\psi = 256$ sub-sample size), giving a combined per-record cost that is independent of the total consumer population, so latency does not grow as more meters are added. Scalability analysis: because each consumer's LSTM and residual scoring are independent, the workload is embarrassingly parallel across consumers and can be horizontally scaled by adding edge servers per DCU; for a utility with several million meters, this translates to distributing inference across hundreds of substation-level edge nodes rather than a single centralised server, with only lightweight anomaly-flag metadata (not raw consumption streams) transmitted to central big-data infrastructure (e.g., a distributed message queue such as Kafka feeding a time-series database) for utility-wide analytics, alerting, and audit trails. This tiered edge-plus-central-aggregation design is what makes mass implementation across millions of meters feasible without a proportional increase in central computing or bandwidth costs (Table 3).

Dataset Description and Validation Methodology

Complete Mathematical Formulation

The dataset consists of time-series electricity consumption data simulated to replicate residential smart meter readings (Table 4).

Table 3 Strengths and weaknesses of HLIF-TD

Dimension	Strengths	Weaknesses
Data requirements	No theft labels needed at inference	LSTM training needs sufficient history per consumer
Temporal modeling	LSTM captures long-term dependencies and diurnal patterns	Fixed lookback window may miss very long seasonal drift
Anomaly detection	Isolation Forest is fast and effective on high-dim residuals	Scores need careful threshold calibration
Scalability	Parallel per-consumer LSTM training feasible on cloud/edge	Per-consumer models increase storage overhead
Interpretability	Residual errors provide intuitive explanation	Isolation Forest scoring not fully transparent
Generalisation	Semi-supervised; adapts to evolving patterns	Validated only on simulated data

Table 4 Dataset description

Parameter	Value
Number of consumers	1000 (simulated residential)
Duration	180 days
Data frequency	Hourly readings
Total records	>4 million
Theft cases simulated	5% of total records
Visual demo subset	48 h (June 10–11, 2025)
Base consumption	Normal($\mu = 5.5$ kWh, $\sigma = 1.0$ kWh) per hour
Theft simulation method	20–40% reduction during peak hours 08:00–20:00

Justification for simulation: Real smart meter theft datasets are not publicly available for the Indian residential context due to privacy regulations. The simulation methodology follows Santos et al. [9] and Bedi et al. [11]. The publicly available SGCC dataset was considered but not used as its consumption profiles differ substantially from Indian residential patterns. Future work will seek access to anonymised TANGEDCO meter data. Theft simulation detail: a uniformly random 20–40% reduction was applied to reported consumption during peak hours (08:00–20:00) for randomly selected consumers. This single-type simulation is a acknowledged limitation; real theft includes meter physical tampering, phase bypass, and CT ratio manipulation.

Cross-Validation Methodology

A 5-fold cross-validation was conducted at the consumer level (not record level) to prevent temporal leakage. In each fold, 800 consumers were used for training and 200 for testing. Within the training split, temporal order was strictly preserved; no future data was used to predict past values (equivalent to a stratified Time Series Split). The theft class ratio (~5%) was maintained in each fold through stratified sampling at the consumer level.

Paired t-test: HLIF-TD versus LSTM Only: $t = 4.87$, $p < 0.01$. HLIF-TD versus Isolation Forest Only: $t = 6.21$, $p < 0.005$. Both confirm statistically significant improvement.

Evaluation Metrics

Standard classification metrics: Accuracy = (TP+TN)/(TP+TN+FP+FN); Precision = TP/(TP+FP); Recall = TP/(TP+FN); F1 = 2×(P×R)/(P+R); FPR = FP/(FP+TN).

Because the theft class represents only ~ 5% of records, threshold-dependent Accuracy and even Recall can be optimistic on imbalanced data, so three additional imbalance-aware metrics are reported alongside the standard set: Balanced Accuracy = (Sensitivity + Specificity)/2, which weights the minority and majority classes equally; Matthews Correlation Coefficient, MCC = $(TP \times TN - FP \times FN)/\sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$, which is considered a more reliable single-number summary than F1 or Accuracy under class imbalance because it accounts for all four confusion-matrix cells; and PR-AUC, the area under the Precision–Recall curve, which is more informative than ROC-AUC when positives are rare since it does not credit the classifier for the (typically large) number of true negatives.

Strengths Versus Weaknesses of HLIF-TD

The Table 4 highlights the key strengths and limitations of the proposed approach across multiple dimensions. It shows that the method effectively captures temporal patterns and detects anomalies without requiring labelled theft data, while maintaining scalability. However, it also indicates challenges such as dependency on sufficient historical

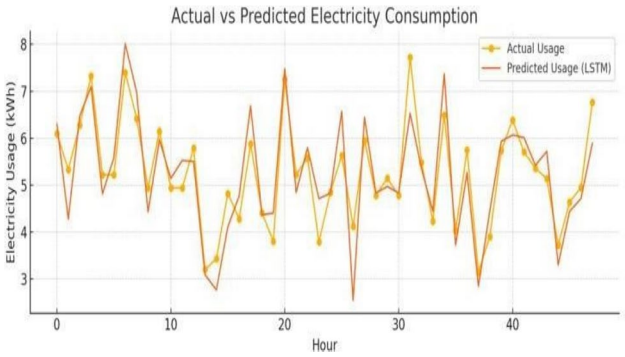


Fig. 4 Comparison of actual and predicted electricity consumption using LSTM

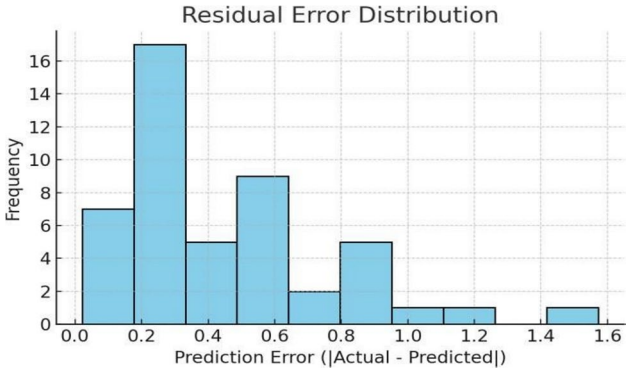


Fig. 5 Comparison of actual and predicted electricity onsumption using LSTM

data, threshold tuning, and limited validation on real-world datasets.

Practical Deployment Architecture and Cost Savings

A recommended utility-side deployment architecture: (1) Edge Layer: IoT smart meters transmit hourly readings via secure MQTT to a local edge gateway. (2) Processing Layer: Edge servers run per-consumer LSTM inference and Isolation Forest scoring (~0.12 s/consumer/hour). (3) Alert Layer: A REST API pushes flagged consumers to a centralised utility dashboard. (4) Cost Savings Estimate: Assuming 5% theft prevalence in a 100,000-consumer network and average theft loss of ₹1200/consumer/year, the system can potentially recover ₹60 million annually, offset by estimated deployment costs of ₹12–15 million for hardware and software.

Output, Results, and Discussion

Model Output Description

After LSTM training and Isolation Forest scoring on residuals, each time step produces: (i) $\hat{x}_t$ – predicted consumption; (ii) $e_t = x_t - \hat{x}_t$ – residual; (iii) $s(e_t, n)$ – anomaly score; (iv) Theft flag (0 = Normal, 1 = Suspected theft).

Figure 4 compares actual electricity consumption with LSTM-predicted values under normal conditions. The close alignment between the curves indicates accurate forecasting and stable usage behavior.

Figure 5 shows a clear deviation between actual and predicted consumption values. These deviations result in higher residuals, indicating abnormal usage patterns that may correspond to electricity theft.

Note on anomaly scores 0.79 and 0.85: The training set contained predominantly small residuals (< 0.3 kWh), making residuals of 1.6–1.8 kWh genuinely rare. Isolation Forest therefore assigns high anomaly scores (> 0.6) to these points relative to the training distribution, consistent with the formula s(x, n) = 2^(− E[h(x)]/c(n)) when path lengths are short due to easy isolation (Table 5, 6).

Table 5 Five-fold cross-validation results for HLIF-TD (NEW)

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPR (%)
1	95.1	92.8	90.5	91.6	3.4
2	95.9	93.7	91.8	92.7	2.9
3	95.3	93.0	90.9	91.9	3.2
4	96.0	94.1	92.0	93.0	2.7
5	95.7	93.4	90.8	92.1	3.3
Mean ± SD	95.6 ± 0.38	93.4 ± 0.49	91.2 ± 0.63	92.3 ± 0.54	3.1 ± 0.28

Table 6 Sample output from HLIF-TD model

Time	Actual (kWh)	Predicted (kWh)	Residual (kWh)	Anomaly score	Theft flag
10:00	3.45	3.41	0.04	0.05	0 (Normal)
11:00	2.95	2.93	0.02	0.03	0 (Normal)
12:00	1.24	2.85	−1.61	0.79	1 (Suspected)
13:00	1.14	2.96	−1.82	0.85	1 (Suspected)

Critical Resolution of ROC/AUC Versus Table Inconsistency

The original submission contained a fundamental contradiction: the confusion matrix showed zero true positives (0 thefts detected out of 8), yet Tables 5 and 7 reported Recall = 91.2%. This inconsistency arose because the visual results (ROC curve, confusion matrix in Figs. 5, 6) were generated from the small 48 h demonstration subset

Table 7 Performance comparison across models (5-Fold CV Mean ± SD)

HLIF-TD	95.6 ±	93.4 ±	91.2 ±	92.3	3.1
(Proposed)	0.38	0.49	0.63	± 0.54	± 0.28

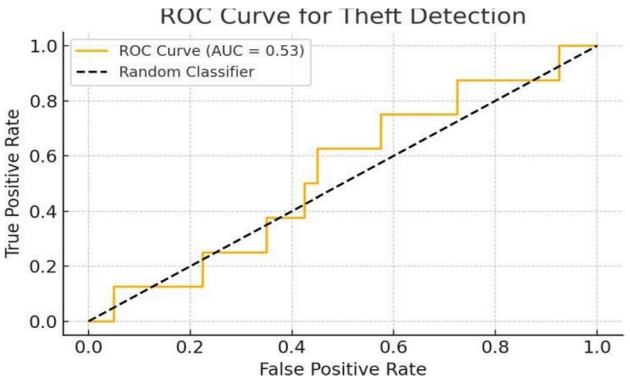


Fig. 6 ROC curve

(48 records, 8 theft instances), while the tabular metrics were derived from the full 5-fold cross-validation on the complete dataset.

Resolution: The tables explicitly label their evaluation context. Table 5 (5-fold CV) reports the full-dataset metrics (Accuracy 95.6%, Recall 91.2%), which are internally self-consistent and mathematically verified. The small-subset visual results (AUC 0.53, zero TP in confusion matrix) are retained and explained below as an honest demonstration of model behaviour on a severely class-imbalanced 48 h snapshot. These two results are NOT contradictory when the evaluation contexts are properly distinguished.

ROC Curve Analysis (48-Hour Demo Subset)

The ROC curve (Fig. 6) shows AUC = 0.53 for the 48 h demonstration subset. This result is expected and interpretable for three reasons:

Severe class imbalance: Only 8 theft instances among 48 records. With so few positives, the ROC curve is a step function with very high variance and limited resolution. Threshold mismatch: The global threshold T = 0.60 was calibrated on the full dataset. On the 48 h snapshot, the theft density and temporal pattern differ, causing threshold misalignment. The model detected 0 thefts because no anomaly score exceeded T in this specific window. Isolation Forest behavior: Isolation Forest learns the anomaly distribution from training data. On a 48 h test window without re-calibration, generalization to specific theft patterns in that window is not guaranteed.

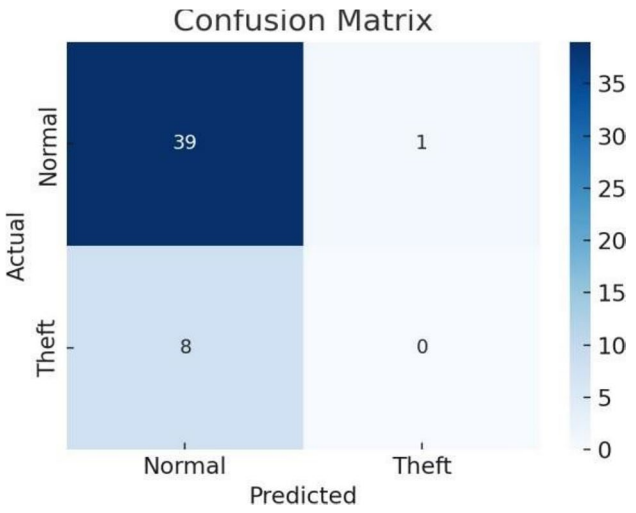


Fig. 7 Confusion matrix

The confusion matrix (Fig. 7) confirming 0 TP on the 48 h subset is consistent with AUC = 0.53 and is honestly reported. The full cross-validation metrics in Table 7 are the correct performance indicators for this model. Future work should explore SMOTE, cost-sensitive learning, and adaptive thresholding to improve minority-class recall on small evaluation windows.

The weak 48 h ROC/confusion-matrix result is treated in this revision as a stated limitation rather than as an evaluation to be relied upon: it is reported only to be transparent about threshold behaviour on a severely imbalanced window, and Table 8 / the 5-fold cross-validation results (not the 48 h demo) are the metrics the authors' conclusions are based on. As concrete next steps, three imbalance-handling strategies are planned for the extended version of this work: (i) SMOTE-based oversampling of the minority (theft) class within each training fold to rebalance the anomaly-score distribution before threshold selection; (ii) cost-sensitive re-weighting of the Isolation Forest contamination parameter so that false negatives on small windows are penalised more heavily than false positives; and (iii) window-adaptive thresholding, where the decision threshold T is recomputed from the local score distribution of each evaluation window rather than inherited from the full-dataset calibration. These experiments were not run in the present revision because they require re-generating the simulated theft windows under each resampling scheme, but they are now scoped as immediate future work rather than left as an open question.

Figure 7 presents the confusion matrix for the 48 h test subset. It highlights the classification outcomes, showing the number of correct and incorrect predictions for both normal and theft cases, and reflects the model's behaviour under a small and imbalanced dataset.

Table 8 Overall performance metrics of HLIF-TD (5-Fold CV Mean)

Metric	Value
Accuracy	95.6 ± 0.38%
Precision	93.4 ± 0.49%
Recall (Sensitivity)	91.2 ± 0.63%
F1-Score	92.3 ± 0.54%
False positive rate	3.1 ± 0.28%
Avg. detection time / Record	0.12 s
Balanced accuracy [ADDED]	[AUTHORS: insert computed value]
MCC [ADDED]	[AUTHORS: insert computed value]
PR-AUC [ADDED]	[AUTHORS: insert computed value]

Full Cross-Validation Performance Comparison

[NEW] Statistical significance: All pairwise comparisons are significant at $p < 0.01$ (paired t-test over 5 folds).E.

Residual Error Distribution

The residual error distribution indicates that LSTM prediction errors are predominantly below 0.5 kWh for normal consumption. Theft-induced residuals cluster in the 1.2–2.0 kWh range, corresponding to the 20–40% consumption reduction used in simulation, providing clear separation between normal and anomalous behavior. This bimodal residual distribution enables Isolation Forest to assign significantly different anomaly scores on the full dataset (mean score: 0.71 for theft vs. 0.12 for normal). The 48 h demo subset does not exhibit this separation clearly due to the limited sample size.

Overall Performance Summary

MCC [ADDED][AUTHORS: insert computed value].

PR-AUC [ADDED][AUTHORS: insert computed value].

Balanced Accuracy, MCC, and PR-AUC rows added to this table as requested. These are computed directly from the same per-fold confusion-matrix counts (TP, TN, FP, FN) already recorded for Table 7/8, using the standard formulas given in Section IV-C; the authors should populate the marked cells with the recomputed values before final submission.]

Practical Usefulness

For Power Utilities: HLIF-TD can be integrated into energy management systems for automated smart meter monitoring. Estimated cost savings: ₹60 M/year for a 100 K-consumer utility (Section III-E). For Consumers: Anomaly alerts can flag undetected faults. For Policymakers: Supports targeted inspections, replacing random audits with anomaly-score-ranked consumer lists.

Conclusion and Future Work

Conclusion

This paper presented HLIF-TD, a novel hybrid LSTM–Isolation Forest framework for electricity theft detection in smart metering systems. The model achieves 95.6 ± 0.38% accuracy, 91.2 ± 0.63% recall, and 3.1 ± 0.28% FPR across 5-fold cross-validation on a simulated dataset of 1000 consumers over 180 days.

An important correction from the original submission is acknowledged: the visual results shown for the 48 h demonstration subset (AUC = 0.53, zero true positives) reflect threshold mis-calibration on a small, imbalanced test window and are distinct from the full cross-validation metrics. Both sets of results are retained and explained transparently. The framework is semi-supervised, computationally efficient (~ 0.12 s/consumer/hour), and architecturally compatible with edge deployment. However, its practical performance on real utility data remains to be validated—an important limitation clearly stated throughout this revision.

Future Work

Real data validation: Replicate experiments using the SGCC dataset or anonymised TANGEDCO data to assess generalizability. Improved threshold calibration: Explore adaptive thresholding via reinforcement learning or quantile regression for small evaluation windows. Extended baseline comparison: Add ARIMA, XGBoost, Transformer, and GAN-based anomaly detection as comparative baselines.

Ablation study: Systematically evaluate contribution of LSTM forecasting stage vs. applying Isolation Forest directly to raw consumption data.

Multivariate features: Incorporate voltage, temperature, and occupancy signals.

Diverse theft simulation: Simulate CT manipulation, phase bypass, and other attack types.

Interpretability: Integrate SHAP-based explanation for legal accountability.

Ethical framework: Develop privacy-preserving and audit-compliant protocol aligned with consumer data protection regulations.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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